

# Dataset Description

## Helmet Detection Object Detection Dataset Report

### 1. Overview

This dataset was created for the task of helmet detection, a binary object detection problem focused on identifying whether individuals riding two-wheeled vehicles are wearing a helmet. Each image is annotated with bounding boxes marking every visible rider, labeled according to helmet usage. The dataset supports the training, validation, and evaluation of object detection models for road-safety and compliance-monitoring applications.

### 2. Dataset Specification

|                           |                                           |
|---------------------------|-------------------------------------------|
| Total Images              | 1,376                                     |
| Total Annotated Instances | 3,618                                     |
| Number of Classes         | 2 (With Helmet, Without Helmet)           |
| Annotation Type           | Bounding box (object detection)           |
| Annotation Format         | YOLO format (.txt) — one file per image   |
| Image Format              | JPG                                       |
| Image Dimensions          | 416 × 416 pixels (standardized)           |
| Split Ratio               | Train 86.1% / Validation 9.2% / Test 4.7% |

### 3. Class Definitions

The dataset defines two mutually exclusive classes, applied per detected rider rather than per image, since a single image may contain multiple riders with different helmet statuses.

| Class              | Instances | Share of Total | Description                |
|--------------------|-----------|----------------|----------------------------|
| With Helmet (0)    | 2,390     | 66.1%          | Rider wearing a helmet     |
| Without Helmet (1) | 1,228     | 33.9%          | Rider not wearing a helmet |

### 4. Dataset Split

The dataset is partitioned into three independent subsets — training, validation, and test — to support proper model development and unbiased evaluation. The training set is used to fit model parameters, the validation set is used for hyperparameter tuning and monitoring generalization during training, and the test set is held out entirely for final performance evaluation.

| Split      | Images | % of Total | With Helmet | Without Helmet | Total Instances |
|------------|--------|------------|-------------|----------------|-----------------|
| Training   | 1,185  | 86.1%      | 2,081       | 1,050          | 3,131           |
| Validation | 127    | 9.2%       | 185         | 115            | 300             |
| Test       | 64     | 4.7%       | 124         | 63             | 187             |
| Total      | 1,376  | 100%       | 2,390       | 1,228          | 3,618           |

#### 4.1 Split Methodology

The split was performed at the image level prior to annotation export, ensuring that no image and its associated annotations appear in more than one subset. This prevents data leakage between training and evaluation and

ensures that reported performance metrics reflect genuine generalization to unseen data. A training-weighted split (86.1% / 9.2% / 4.7%) was adopted to maximize the number of examples available for model learning, while retaining a sufficient validation set for reliable tuning and a held-out test set for final, unbiased evaluation.

## 5. Class Distribution and Balance

Across the full dataset, the class distribution is moderately imbalanced, with “With Helmet” instances outnumbering “Without Helmet” instances at an approximate ratio of 1.95 to 1. This imbalance is consistent across all three splits and is not expected to severely bias model training, though it is noted here as it may influence per-class precision and recall during evaluation.

## 6. Coverage of Visual Variation

To support robust model generalization, the dataset captures variation across several key visual factors, summarized below.

| Factor             | Coverage in Dataset                                                                     |
|--------------------|-----------------------------------------------------------------------------------------|
| Background         | Urban streets, plazas, open roads, and mixed pedestrian/traffic environments            |
| Lighting           | Natural daylight conditions with varying brightness and shadow levels                   |
| Viewing Angle      | Front-facing, side-profile, and rear viewing angles of riders                           |
| Occlusion          | Partial occlusion from other riders, vehicles, and background objects                   |
| Scale              | Riders captured at varying distances, producing small, medium, and large bounding boxes |
| Subjects per Image | Single riders and multiple riders within the same frame                                 |

## 7. Annotation Format

Each image is accompanied by a corresponding label file in YOLO format. Every line in a label file represents a single bounding box using normalized coordinates:

**class\_id x\_center y\_center width height**

All coordinate values are normalized relative to image width and height (range 0 to 1). The class\_id corresponds to the class index defined in Section 3 (0 = With Helmet, 1 = Without Helmet).

- Example (With Helmet): 0 0.4017 0.2650 0.1064 0.1550
- Example (Without Helmet): 1 0.5303 0.4105 0.0895 0.1863

## 8. Annotation

All images in the dataset were annotated with bounding boxes prior to use, assigning each visible rider one of the two defined classes. Annotation quality was verified along three criteria: bounding box accuracy, label correctness, and completeness (no missing objects).

### 8.1 Annotation Summary

|                  |       |
|------------------|-------|
| Number of Images | 1,376 |
|------------------|-------|

|                                        |                                 |
|----------------------------------------|---------------------------------|
| Number of Labels (Annotated Instances) | 3,618                           |
| Number of Classes                      | 2 (With Helmet, Without Helmet) |

## 8.2 Completeness Check

A programmatic check was performed to confirm that every image in the dataset has a corresponding, non-empty label file. No images were found to be missing annotations, and no label files were found to be empty.

- Images missing a label file entirely: 0
- Label files present but containing zero bounding boxes: 0
- Result: 100% of images have at least one valid annotated instance

## 8.3 Accuracy Verification

Bounding box accuracy and label correctness were verified through visual spot-checking. A random sample of annotated images was rendered with their bounding boxes and class labels overlaid, then manually inspected to confirm that boxes tightly enclose the intended object and that class labels (With Helmet / Without Helmet) correctly match the visual content. The sample below illustrates this verification process.

